

What LLM Forecasters Reveal When You Ask Them About Their Uncertainty: A Per-Family, Cross-Corpus Study

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Abstract

Five LLM families (Claude Opus 4.7, GPT-5.5 and GPT-5.4-mini via Codex CLI, Gemini 2.5 Flash, DeepSeek Chat) are asked the same binary forecasting questions across two corpora that differ on four confounded axes (length, structural complexity, pre-training overlap, public-prior availability). We elicit uncertainty through multiple primitives in the same prompts — a worry scalar, a bid-ask spread, an interval, a reasoning trajectory, an outside-view reference class — and resolve every claim through one of three legal verdicts (**h1_supported**, **h0_kept**, **inconclusive_underpowered**) with sample sizes pre-computed via Fisher- z power analysis. The headline result is that LLM uncertainty does not decompose into a single signal. We document six findings: (1) a worry scalar elicited alongside the forecast is Bayesian-coherent tail risk at $N=590$ pool level, validating the basic primitive; (2) the per-model worry-Brier relationship is direction-flipped between families and replicates across three independent pilots ($N=264$); (3) the rationale content and the point estimate decouple ($N=320$ analytic) — LLMs surface failure reasoning but do not recompute their probability from it; (4) a novel meta-uncertainty primitive (bid-ask spread, with no precedent in the LLM-calibration literature) tracks realized error at $\rho = +0.69$ ($p=0.013$) for two of five families on the specialist corpus, with cross-corpus replication underpowered at $N=42$; (5) three uncertainty channels (worry, spread, trajectory variance) are statistically independent — no universal best uncertainty primitive across families; (6) Tetlock-style frequency framing improves GPT-5.4-mini on the specialist corpus ($\Delta = -0.118$, $p=0.013$) but does NOT replicate on $N=42$ public-domain questions, with three of five families’ Brier-deltas **h0_kept** at the ± 0.05 equivalence bound. The audit pipeline also retracted four prior-program “null” findings to **inconclusive_underpowered** and five cross-corpus claims to corpus-specific. We propose a refined mechanism for the worry-scalar sign-flip: families share a topic-trigger (the same content topics are flagged high-worry across LLMs), but per-family accuracy on those flagged topics is heterogeneous — the apparent inversion is a topic-quality $\times$ per-family-accuracy interaction, not a per-family worry-emission bias. The deployable recipe is a per-family elicitation policy rather than averaged probability across an LLM panel.

1 Introduction

Recent LLM-forecasting work reports impressive multi-agent-ensemble accuracy [3] and verbalized-confidence calibration [5, 2]. What is uncommon: per-family heterogeneity in calibration responses, cross-corpus replication at proper power, and power-aware verdict resolution. We test a five-model panel on two corpora that span the four confounded axes a real deployment will face (long-and-alien vs short-and-familiar), apply a pre-registration discipline before every pilot, and report what survives.

Two operational claims motivate the writeup. *First*, the field is converging on a recognizable positive-result template at the pool/aggregate stratum that masks where the per-model story actually

lives. *Second*, in our own program we re-examined every prior “no effect” claim under a Fisher- z power-aware verdict resolver: 8 of 12 turned into `inconclusive_underpowered` and 5 cross-corpus claims retracted to corpus-specific. The audit ITSELF is part of the methodological contribution because the same overcalling pattern likely affects the field’s existing $N=5\text{--}20$ cross-corpus claims.

The contributions, in the order they appear:

1. The worry-scalar is a Bayesian-coherent tail-risk signal at pool level, foundational to everything that follows (§3).
2. Per-family worry-Brier direction is split between families and replicates (§4).
3. Cognitive content and calibration shift are decoupled, well-powered analytic (§5).
4. Bid-ask spread is a novel deployable runtime calibration warning on the specialist corpus (§6).
5. Three uncertainty channels are statistically independent: per-family best-channel routing required (§7).
6. Frequency framing transfers in a corpus-specific, family-specific way (§8).
7. Refined mechanism for the F36 sign-flip: shared topic-trigger $\times$ per-family accuracy (§9).

The paper closes with an honest scope statement (§11) and the reproducibility plan (§12).

2 Setup

2.1 Five LLM families

Claude Opus 4.7 (Anthropic, via Claude Code CLI), GPT-5.5 and GPT-5.4-mini (OpenAI, via Codex CLI), Gemini 2.5 Flash (Google GenAI API), DeepSeek Chat (DeepSeek API). All API calls temperature=0 unless explicitly noted. All paths audited for web-search-tool access; all prompts include an explicit “do NOT use web search” instruction.

2.2 Two corpora, four confounded axes

We use a *specialist corpus* of $N=15$ bespoke questions about a private research program (formal-proof tactic combinations, gated state transitions in a substrate-specific apparatus, operator-confirmed forecasting contracts), and a *public-domain corpus* of $N=42$ contracts from prediction markets (Manifold, Polymarket), stock-close thresholds, and ETF moves.

Axis	Specialist ($N=15$)	Public-domain ($N=42$)
LLM pre-training overlap	Effectively zero	Substantial
Question length	247 chars (35 words)	99 chars (17 words)
Question structure	Multi-conditional, code/math symbols, parenthetical embedding	Atomic, simple syntactic structure
Implicit base rate	Ill-defined (bespoke operations)	Public-market priors well-trained

The four axes are confounded in the current data. Specialist questions are long-complex-alien-no-public-prior *together*; public-domain are short-simple-familiar-with-public-prior *together*. Our

cross-corpus result (§8) is therefore evidence that *at least one* of these axes matters; a 4-cell de-confounded corpus design that breaks the confound is the highest-priority methodological follow-up (§11).

The specialist corpus is operator-private; the reproducibility plan (§12) addresses this through a sanitized release with neutral identifiers and a parallel public-but-niche-academic corpus matching the four-axis profile.

2.3 Pre-registration and power-aware verdicts

Every pilot is pre-registered in a SQLite `pre_registrations` table before launch with hypothesis, null, target effect size, n_{required} (auto-computed via Fisher- z at $\alpha=0.05$ two-tailed, 80% power, Spearman correction +6%), falsifiers, and pass-gate. Resolution uses three legal verdicts:

- `h1_supported`: 95% CI on observed ρ excludes 0.
- `h0_kept`: 95% CI wholly within $(-\text{target}, +\text{target})$; data rules out a meaningful effect.
- `inconclusive_underpowered`: observed $|\rho|$ below detectability at the run N ; CI wide.

Brier-delta tests use paired-contract sign-flip permutation with a 90% bootstrap CI plus a BIC-approximation Bayes factor. “No effect” claims also require TOST equivalence at a pre-stated bound. Power thresholds at $\alpha=0.05$ / power= 0.80: $N=15 \Rightarrow |\rho| \geq 0.68$; $N=42 \Rightarrow 0.43$; $N=91 \Rightarrow 0.30$.

3 Foundation: the worry scalar is Bayesian-coherent tail risk

The basic uncertainty primitive used throughout this paper is `tail_insurance_premium` (worry scalar, 1–100, requested in the same prompt as the binary forecast). This primitive was validated in a separate within-program pilot before any of the downstream analyses: at pool level $N=590$ on an independent cross-pool forecast corpus, the worry scalar is a Bayesian-coherent tail-risk signal — more-worried forecasts ARE more likely to be wrong, in expectation. This is the only well-powered *positive* pool-level validation of a verbalized-uncertainty primitive in our program; everything that follows uses worry as a base channel, and the per-model decomposition in §4 shows the pool-level signal is the average of opposite per-family signs.

4 Per-family worry-Brier direction split, three-pilot replication

For each LLM family we compute the rank correlation between the worry score and the realized squared error (Brier) of the forecast that the worry accompanied. Pilot v21 ($N=164$ premium-rowed calls), v22b ($N=85$), and v22d ($N=115$) replicate this analysis independently, cumulative $N=264$:

- Claude Opus 4.7: $\rho < 0$ — most worried when most right (anti-Dunning-Kruger).
- GPT-5.5: $\rho < 0$ — same direction.
- GPT-5.4-mini: $\rho > 0$ — direction-sensible (worried when wrong).

Per-pilot detectable $|\rho|$ at 80% power is ≈ 0.30 for $N=85$; observed $|\rho|$ values span 0.31–0.56 across families, clearing detectability for Claude and GPT-5.5 individually. Cross-pilot Stouffer Z meta-analysis combines the three pilots at combined $p < 0.01$. **This is the most robust per-model finding in the program.**

A natural reviewer hypothesis is that the sign-flip is mediated by task complexity: complex tasks produce longer rationales; longer rationales correlate with both higher stated worry AND lower Brier (more thorough reasoning); the apparent inversion is an artifact of this joint mediation. We test the mediator directly on $N=45-91$ per family. None of the five families exhibits the predicted pattern of “long rationale $\rightarrow$ high worry AND long rationale $\rightarrow$ low Brier.” Claude’s $\rho(\text{rat_length}, \text{tip}) \approx 0$ (claude does not worry more about long rationales); Gemini’s $\rho(\text{rat_length}, \text{Brier})$ is POSITIVE, the opposite of the reviewer’s predicted direction. **The task-complexity-via-rationale-length mediator is rejected.**

The refined mechanism we propose is given in §9 after §5 provides the upstream decoupling result.

5 Cognitive content and calibration shift are decoupled

Across pilots v22, v22d, and v24 (pooled $N=320$ analytic), we measure the within-family Spearman correlation between the count of failure-mode words in the rationale and the per-row Brier:

Family	N	$\rho(\text{failure-mentions}, \text{Brier})$
Claude Opus 4.7	≈ 60	+0.27
GPT-5.5	≈ 60	+0.18
GPT-5.4-mini	≈ 60	+0.21
Gemini 2.5 Flash	60	− 0.045
DeepSeek Chat	59	+ 0.049

At $N=60$ per family the detectable $|\rho|$ at 80% power is ≈ 0.36 . Gemini and DeepSeek’s near-zero ρ values fall within the **h0.kept** band at $|\rho| < 0.20$. The model produces failure-mode reasoning text on demand — cognitive shift is real — but does not recompute the point estimate from that content. This is the cleanest mechanism evidence in the program and connects to the chain-of-thought-faithfulness lineage [1]; the original trio shows weak positive correlation consistent with partial propagation rather than full decoupling.

6 Bid-ask spread: a novel meta-uncertainty primitive

Instead of asking the model for one probability, we elicit three in the same prompt: the highest probability at which it would BUY a YES contract, its point estimate, and the lowest probability at which it would SELL YES. The spread (sell − buy) is the model’s self-reported uncertainty about its own probability — a Knightian bracket distinct from the unidirectional worry scalar.

Family	Mean spread	$\rho(\text{spread}, \text{Brier})$	95% CI (Fisher- z)	Verdict
Claude Opus 4.7	0.21	+0.49	[−0.03, +0.80]	CI barely touches 0
GPT-5.5	0.24	−0.05	[−0.55, +0.47]	inconclusive
GPT-5.4-mini	0.26	+ 0.69	[+ 0.28 , + 0.89]	h1_supported , $p=0.013$
Gemini 2.5 Flash	0.21	+0.40	[−0.15, +0.75]	inconclusive
DeepSeek Chat	0.26	+ 0.69	[+ 0.27 , + 0.89]	h1_supported , $p=0.013$

Pre-registered pass-gate met: two of five families clear $\rho > 0.3$ with non-zero-crossing CI at $N=15$ specialist. No family shows the wrong sign. To our knowledge no prior LLM-forecasting paper has elicited bid-ask spread as an uncertainty primitive. The market-microstructure framing

forces the model into a more demanding cognitive task than the scalar worry, which we conjecture is exactly why it produces a more honest meta-uncertainty signal (the elaboration is supported by the channel-orthogonality result in §7).

Cross-corpus failure-to-replicate. On the public-domain $N=42$ corpus, all per-family CIs cross zero (detection threshold $|\rho| \geq 0.43$ at 80% power for $N=42$). The point-estimate ρ values for GPT-5.4-mini (-0.11) and DeepSeek (-0.07) flip sign. Either (a) the primitive is corpus-specific consistent with the four-axis confound, or (b) $N=42$ remains underpowered to confirm a smaller-but-real positive effect. To distinguish requires $N \geq 91$ public-domain. The honest framing is *specialist-corpus positive with explicit cross-corpus failure-to-replicate*.

7 LLM uncertainty channels are statistically independent

Pooling three specialist-corpus full pilots (worry-scalar baseline, bid-ask spread, reasoning-trajectory variance) on the $N=75$ rows where all three are observed, the pairwise Pearson correlation matrix among the three uncertainty channels:

	worry scalar	bid-ask spread	trajectory variance
worry scalar	1.00	+0.08	+0.17
bid-ask spread	+0.08	1.00	+0.03
trajectory variance	+0.17	+0.03	1.00

The off-diagonals are all below 0.20. PCA would not produce dimensionality reduction. Each primitive is measuring something genuinely different about how a model expresses uncertainty.

Per-family $\rho(\text{channel}, \text{Brier})$ reveals that each model family has a different “best channel”:

Family	$\rho(\text{worry}, \text{Brier})$	$\rho(\text{spread}, \text{Brier})$	$\rho(\text{traj_var}, \text{Brier})$	Best channel
Claude Opus 4.7	-0.38 (sign-flipped)	+0.02	+0.08	worry (with sign flip)
GPT-5.5	+0.07	+0.12	-0.18	weak across all
GPT-5.4-mini	-0.18	+0.55	-0.22	bid-ask spread
Gemini 2.5 Flash	+0.10	+0.10	+0.25	trajectory variance
DeepSeek Chat	-0.26 (sign-flipped)	+0.29	-0.10	bid-ask spread

There is no universal best uncertainty primitive across LLMs. The deployable recipe is a per-family elicitation policy: identify the best channel per family via a pre-registered pilot, then use that channel as the runtime calibration warning. Per-primitive per-family single-channel R^2 on the same-pilot Brier is 10–26% for the families where the channel works (Codex-mini’s bid-ask explains 26% of v27a Brier; DeepSeek’s 19%); multi-channel R^2 on a single forecast remains pending the in-flight v28a pilot that elicits all channels in one prompt.

Scope of the orthogonality claim. The result is at the *expression* level. A reasoning-decomposition claim would require mechanistic interpretability tooling — logit lens, attention probing, causal intervention on chain-of-thought spans — which we do not use. Our primitives are black-box behavioral probes of uncertainty expression; the deployment implication (per-family routing) is unchanged but the mechanistic-reasoning claim is left for white-box work.

8 Frequency framing transfers in a corpus-specific way

Tetlock-style “imagine 10 similar contracts; how many resolve as success?” [4] framing on the specialist corpus ($N=15$ each, paired permutation vs the same-corpus baseline):

Family	Δ -Brier	95% CI	p (perm)	Verdict
Claude Opus 4.7	−0.006	[−0.05, +0.03]	0.85	inconclusive
GPT-5.5	+0.014	[−0.06, +0.09]	0.74	inconclusive
GPT-5.4-mini	−0.118	[−0.21, −0.04]	0.013	h1_supported
Gemini 2.5 Flash	−0.118	[−0.28, +0.03]	0.19	inconclusive
DeepSeek Chat	+0.115	[−0.02, +0.28]	0.18	inconclusive

On the public-domain $N=42$ corpus:

Family	Δ -Brier	95% CI	Verdict at bound 0.05
Claude Opus 4.7	−0.003	[−0.030, +0.028]	h0_kept
GPT-5.5	+0.005	[−0.012, +0.023]	h0_kept
GPT-5.4-mini	+0.003	[−0.042, +0.053]	borderline
Gemini 2.5 Flash	+0.052	[+0.000, +0.112]	not h0_kept ($p=0.071$)
DeepSeek Chat	−0.001	[−0.031, +0.030]	h0_kept

GPT-5.4-mini’s specialist-corpus improvement does NOT replicate on the public-domain corpus. Direction goes from −0.118 to +0.003. For 3 of 5 families the public-corpus Brier-delta is **h0_kept** at bound ± 0.05 — the data actively rules out a 5%-Brier effect on public-domain questions. Sub-source stratification within the public corpus reveals heterogeneity the aggregated null masks: Gemini on Polymarket-style questions trends $\Delta = +0.138$ ($p=0.14$); DeepSeek shows opposite-sign effects on Manifold-bulk vs Polymarket; GPT-5.4-mini is genuinely corpus-specific (every sub-source $\Delta \in [-0.002, +0.007]$, no false-negative from aggregation).

The clean claim. Frequency framing’s effect on LLM forecasters is corpus-specific and model-specific: it improves GPT-5.4-mini on specialist apparatus questions only. On public-domain real-world questions, no model shows a $\geq 5\%$ Brier improvement, and the per-source direction within “public-domain” depends on the question category. Pre-registration prevented us from over-stating the cross-corpus claim as a general failure-of-transfer; the specific finding is the corpus-dependent transfer.

9 Refined mechanism for the worry-scalar sign-flip: shared topic-trigger $\times$ per-family accuracy

A reviewer-flagged concern is that the affect-proxy hypothesis (§4 early framing) was still just a hypothesis. We ran four mechanism probes on existing within-program data to disambiguate.

Probe 1. Within-family rank correlation between (across-family p -variance per contract, used as a proxy for intrinsic question difficulty) and (that family’s mean worry on that contract). GPT-5.4-mini shows $\rho = -0.43$, 95% CI [−0.69, −0.07], **h1_supported negative** — the opposite direction from a “question-difficulty confound” mediator. The mediator hypothesis is rejected.

Probe 2. Stratified $\rho(\text{worry}, \text{Brier})$ by question difficulty (hard half vs easy half) and by point-estimate extremity (uncertain $p \in [0.4, 0.6]$ vs confident). Claude’s inversion lives in the EASY-contracts subset only ($\rho_{\text{EASY}} = -0.19$, $\rho_{\text{HARD}} = +0.03$); GPT-5.5’s inversion lives in the

UNCERTAIN- p subset only ($\rho=-0.31$, $n=21$). The sign-flip is family-specific AND subspace-specific; no single uniform mechanism.

Probe 3. Lexical log-likelihood-ratio of distinguishing tokens between high-worry ($\text{tip}\geq 70$) and low-worry ($\text{tip}\leq 30$) rationales per family. Under affect-proxy, high-worry rationales should be enriched in CAREFUL (*analyze, consider*) or UNCERTAIN (*unsure, difficult*) words. We observe instead that the dominant distinguishing terms are *specific content-topic words* — *tactic, convolution, measurability, singularity* — consistent with a topic-trigger rather than affect-proxy hypothesis.

Probe 4. Cross-family worry agreement: do families AGREE on which contracts get high worry? Pairwise Spearman correlation between each pair of families’ mean-worry-per-contract gives Claude $\leftrightarrow$ GPT-5.5 $\rho=+0.40$ (h1_supported), GPT-5.5 $\leftrightarrow$ GPT-5.4-mini $\rho=+0.41$ (h1_supported), GPT-5.4-mini $\leftrightarrow$ DeepSeek $\rho=+0.64$ (h1_supported). **Families share a topic-trigger signal: they pick the same contracts as high-worry.**

Probe 5. $\rho(\text{worry}, \text{wallclock_seconds})$ per family — does the model “think longer” on high-worry rows? All families show $|\rho|\leq 0.25$ with CIs crossing zero. Worry is NOT a compute-effort proxy.

Probe 6 — mean-regression within row. For each row we measure $\rho(\text{worry}, |p_{\text{success}} - 0.5|)$ per family. Claude: $\rho = -0.39$ 95% CI $[-0.55, -0.20]$, h1_supported negative. Codex-mini: -0.59 $[-0.72, -0.41]$, h1_supported. DeepSeek: -0.35 $[-0.58, -0.06]$, h1_supported. GPT-5.5 and Gemini both null. **Three of five families moderate their probability when emitting high worry.**

Probe 7 — per-contract accuracy alignment. Rank contracts by claude’s mean worry. Compare mean Brier per family in the top-quartile (claude-high-worry contracts) vs bottom-quartile (claude-low-worry contracts):

Family	Top-quartile Brier	Bottom-quartile Brier	Diff
Claude Opus 4.7	0.181	0.392	−0.211
GPT-5.5	0.199	0.224	−0.025
GPT-5.4-mini	0.160	0.219	−0.060

Claude is half as wrong on its high-worry contracts. The other families show small or no asymmetry.

Probe 8 — RLHF hedging mediator (rejected). $\rho(\text{hedging-vocabulary density}, \text{Brier})$ is POSITIVE for Claude (+0.15) and GPT-5.4-mini (+0.31) — opposite the direction of their worry-Brier inversion. Hedging-language is not the mediator.

Converged synthesis: a two-mechanism interaction. The F36 sign-flip emerges from the simultaneous presence of two distinct mechanisms:

1. *Worry-triggered mean-regression.* High worry $\rightarrow$ less-extreme p on the same row (Probe 6). Three of five families show this at h1_supported.
2. *Per-family accuracy alignment with the trigger topics.* The shared topic-trigger (Probe 4) flags long-structured apparatus contracts as high-worry. Claude is GENUINELY BETTER on those contracts (Probe 7, diff -0.21); GPT-5.4-mini is not (diff -0.06).

Claude has BOTH (1) and (2): moderation lands close to truth on contracts where claude is accurate $\rightarrow$ inverted $\rho(\text{worry}, \text{Brier})$ (the F36 signature). GPT-5.4-mini has (1) but not (2): moderation doesn’t save it on topics it’s weaker on $\rightarrow$ direction-sensible $\rho(\text{worry}, \text{Brier})$. GPT-5.5 has neither cleanly at the pooled-across-conditions level, and its F36 inversion (documented at the per-pilot positive-magnitude-balance-off cell, $N=85$) is condition-specific.

The mechanism is identified at evidence-summary grade for Claude (the most cleanly-inverted family). It is partial for the other inverted families. White-box activation-level probing would directly test which activations drive worry emission versus probability emission; we have only the black-box behavioral signature. The deployment implication is unchanged — the worry scalar by itself is not a competent calibration signal across families — but the routing recipe is now mechanistically explained: per-(family, corpus) routing is needed because mean-regression is universal-but-variable, while per-family topic-accuracy alignment is what gives the worry-Brier relationship its sign.

10 Multi-channel R^2 and a NEW channel: self-predicted Brier interval

The orthogonality result (§7) said the three channels are statistically independent. A natural follow-up: how much of forecast Brier do they jointly explain when measured *on the same forecast*? Earlier we could only estimate single-channel R^2 per primitive (5–26%) because each primitive was elicited in a separate pilot with its own forecast. The v28a primitive elicits four channels in ONE prompt: bid-ask spread, scalar worry, AND the lower and upper bounds of the LLM’s self-predicted Brier interval (a previously-unused continuous channel).

Multi-channel OLS R^2 on same-forecast Brier:

Corpus	Family	N	Best single-channel R^2 (which)	Multi-channel R^2 (4 chans)	adj R^2
Specialist	Claude Opus 4.7	15	0.05 (spread)	0.09	(overfit at $N=15$,
Specialist	GPT-5.5	15	0.20 (b_width)	0.39	0.15
Specialist	GPT-5.4-mini	15	0.63 (b_width)	0.66	0.53
Specialist	Gemini 2.5 Flash	15	0.47 (b_mid)	0.90	0.86
Specialist	DeepSeek Chat	15	0.24 (b_mid)	0.26	(overfit)
Public	Claude Opus 4.7	42	0.14 (b_mid)	0.20	0.12
Public	GPT-5.5	42	0.29 (spread)	0.57	0.52
Public	GPT-5.4-mini	42	0.08 (tip)	0.16	0.07
Public	Gemini 2.5 Flash	42	0.24 (tip)	0.31	0.23
Public	DeepSeek Chat	42	0.22 (spread)	0.28	0.20

Two new claims emerge.

First, *multi-channel R^2 is substantive at proper N* . GPT-5.5 on the public-domain corpus reaches adj R^2 0.52 with four channels; Gemini on the specialist corpus reaches 0.86 (subject to overfit caveat at $N=15$). The orthogonality result (§7) and this multi-channel result are consistent — the channels are independent AND informative.

Second, *a new strongest channel emerges: the LLM’s self-predicted Brier interval*. v28a asked each model to emit a 90% CI on its OWN Brier (`predicted_brier_lo`, `predicted_brier_hi`) alongside the forecast. The interval midpoint (`b_mid`) and width (`b_width`) form a continuous self-predicted-Brier channel. For GPT-5.4-mini specialist, `b_width` ALONE explains 63% of Brier variance ($\rho = -0.78$); for Gemini specialist, `b_mid` ALONE explains 47% ($\rho = +0.63$). These are LARGER single-channel R^2 than spread or worry produced on the same data. To our knowledge no prior LLM-calibration paper has elicited self-predicted-Brier as a continuous channel.

The sign-flip continues to migrate across channels per (family, corpus): GPT-5.4-mini `b_width` is inverted on specialist ($\rho = -0.78$); Gemini `b_mid` is direction-correct on specialist ($\rho = +0.63$); GPT-5.5 `tip` is direction-correct on public ($\rho = +0.44$) versus its known specialist-corpus inversion. The two-mechanism interaction (§9) extends to these channels: per-family accuracy patterns interact with each newly-elicited channel, producing different sign-flips per (family, channel, corpus) cell.

Saturation curve. The current 4-channel R^2 is not the ceiling. Additional primitives (interval calibration with explicit coverage, reference-class outside view, meta-channel routing, pre-mortem-with-causes) are queued. The R^2 -vs-channel-count saturation curve is the right summary plot for the deployment recipe; pre-registered as the next analytical pass.

11 What this paper does NOT establish

- It does NOT show that cognitive-debiasing interventions never transfer to LLMs. Four of five tested are **inconclusive_underpowered** at our specialist-corpus N , not falsified.
- It does NOT establish that GPT-5.4-mini’s specialist-corpus frequency-framing improvement would survive cross-corpus replication at proper power. $N \geq 91$ public-domain is required.
- It does NOT generalize beyond five model families and two corpus classes. The specialist corpus is operator-private.
- It does NOT decompose LLM *reasoning* mechanistically. The channel-orthogonality claim (§7) is at the *expression* level; a reasoning-decomposition claim would need white-box mechanistic interpretability.
- The four-axis confounding between corpora means the cross-corpus result is evidence that at least one axis matters, not yet a clean attribution. The 4-cell de-confounded corpus is the most important methodological follow-up.
- Bid-ask spread cross-corpus is currently underpowered; the direction-shift is suggestive but not conclusive at $N=42$.
- The F36 sign-flip mechanism is provisional (§9 multi-probe synthesis); activation-level probing and a topic-matched corpus would settle it.

12 Reproducibility

All pilot calls persist as JSONL files and are ingested into a SQLite database at `analytics/public/calibration/for`. Pre-registrations and verdict resolutions are in the same database. The reusable general-purpose statistics module is at `src/ztare/experiment_stats.py` (power calculator, bootstrap CI, paired permutation, Fisher- z Spearman, TOST, BH-FDR, power-aware verdict, BIC Bayes factor, reproducibility hash). Forecasting-specific wrappers are in the per-project `calibration_stats.py`. Every reported finding reproduces via CLI: `calibration_stats.py finding F54, brier-ci --primitive baseline --corpus internal, spearman --pilot v27a_full__internal --x spread, etc.`

Specialist-corpus reproducibility. The specialist corpus is operator-private. Remediation paths: (a) sanitized release with operator-private identifiers replaced by neutral tags (e.g., “Operation X tactic Y on row class Z ”), preserving the four-axis structural profile; release pending operator clearance. (b) Parallel public alien-domain corpus drawn from public-but-niche academic sub-fields (specific Lean mathlib lemma classes; obscure subfields of partial differential equations; verifiable but low-traffic historical-event resolutions) matching the specialist corpus’s four-axis profile, intended for direct third-party replication of F54 and F56.

References

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